

DATA MINING ASSIGNMENT-1

Objective

Use the provided CSV dataset to perform data visualization and generate meaningful business insights. Apply appropriate visualization techniques to analyse sales performance, trends, and category-level behaviour.

Dataset: CSV data on Warehouse and Retail Sales.

Dimension of data = 307645 rows x 9 columns

Pre-processing:

The dataset required minimal pre-processing. A new “Date” variable was created by combining the YEAR and MONTH fields to enable time-series analysis. Additional features such as Year, Quarter, and Month were extracted from the data itself. Data aggregation was performed to compute total sales across different time intervals and product categories. The dataset was also checked for missing values and data type consistency before performing visualization and analysis.

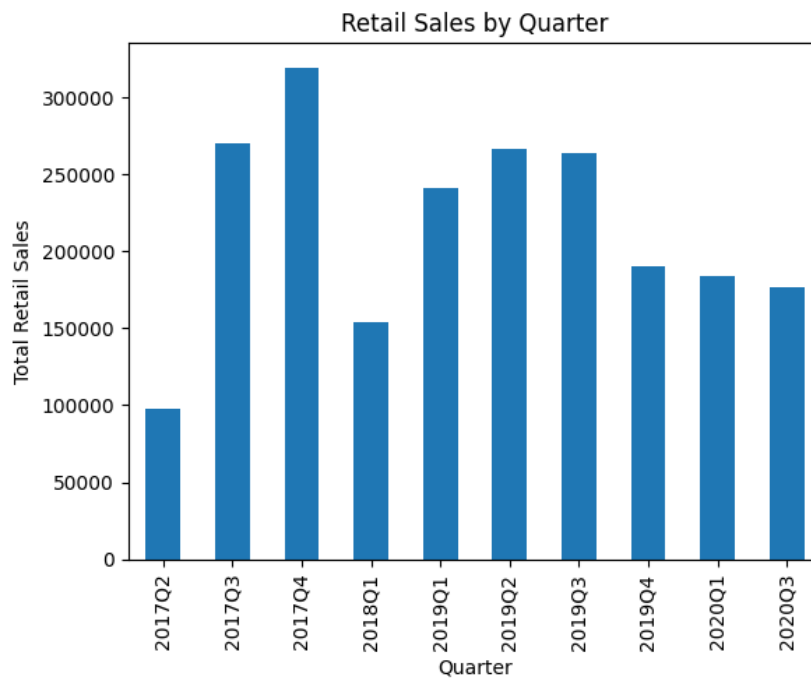
TASK -1

Total Retail Sales over Time

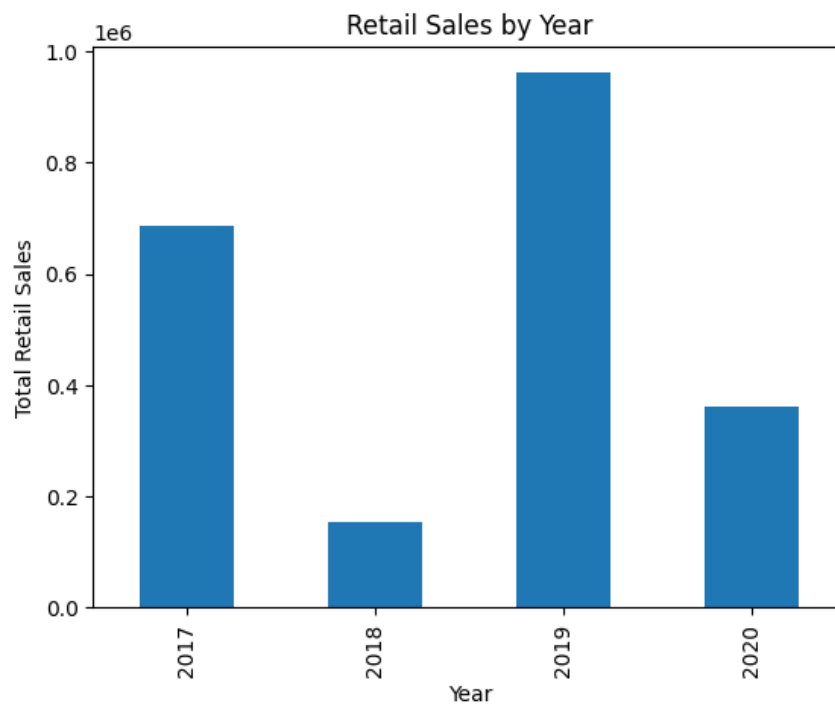
- Visualize total retail sales on a monthly, quarterly, and yearly basis.



Insight : Total Retail Sale is highest in the month of July and lowest in the month of April



Insight : Total Retail Sale is highest in the quarter 4 of 2017 and lowest in the quarter 2 of 2017



Insight : Total Retail Sale is highest in the year 2019 and lowest in the year 2018

• **Identify seasonal patterns, growth trends, and decline periods.**

From the bar chart of Retails sales by month,

- **Seasonal pattern** -> Sales are strongest in July followed by September and January

- **Growth trend** -> Retail sales show a growth trend from the second quarter toward the middle of the year, indicating increased consumer demand during this period.
- **Decline period** -> Sales decline sharply after peak periods, suggesting reduced consumer spending during certain months

From the bar chart of Retail sales by quarter,

- **Seasonal pattern** – Q4 2017 has highest sales followed by a sudden decline. This may indicate a holiday driven demand
- **Growth trend** -> Retail sales shows a sudden growth from Q2 2017 to Q3 2017, showing increased consumer demand for product
- **Decline period** -> Sharp decrease from Q4 2017 to Q1 2018. This sudden decline maybe due to lesser purchase by consumers after holiday period.

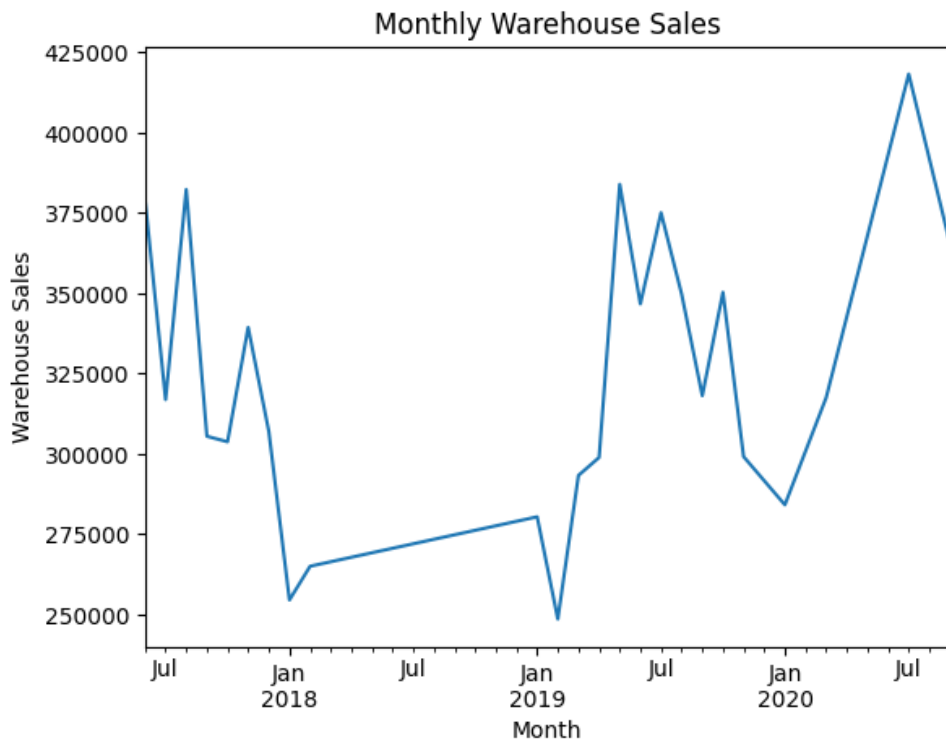
From the bar chart of Retail sales by year,

- **Seasonal pattern** – 2019 has highest sales. This may be because of Covid-19 and lockdown.
- **Growth trend** -> Retail sales shows a sudden growth from 2018 to 2019, showing increased consumer demand for product. This may be due to Covid-19 and lockdown.
- **Decline period** -> Sharp decrease from 2019 to 2020. This sudden decline may be due to back to normal from covid scenario.

TASK – 2

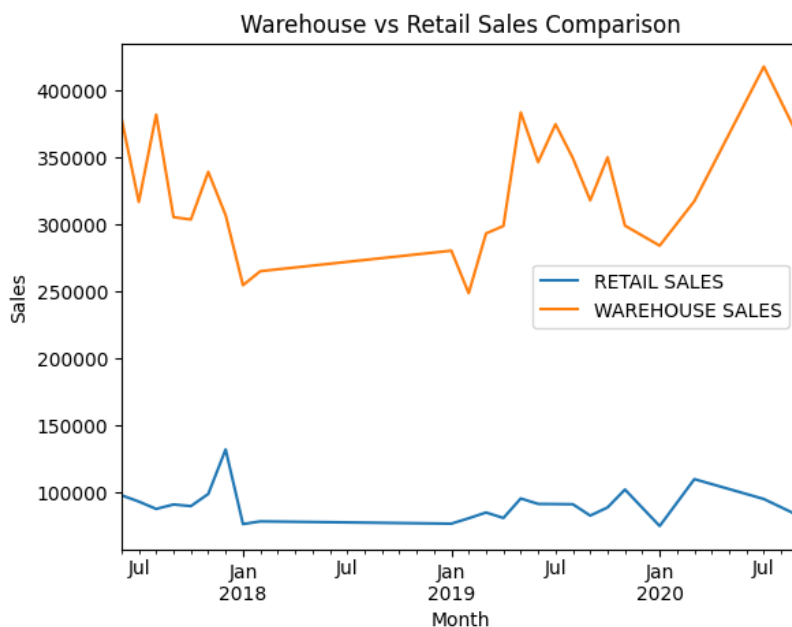
Warehouse Sales Over Time

- **Analyze warehouse sales trends over time.**



Insight : Maximum sales happened in July 2020. Minimum in Jan 2019

- Compare warehouse movement with retail sales to understand supply-demand alignment.

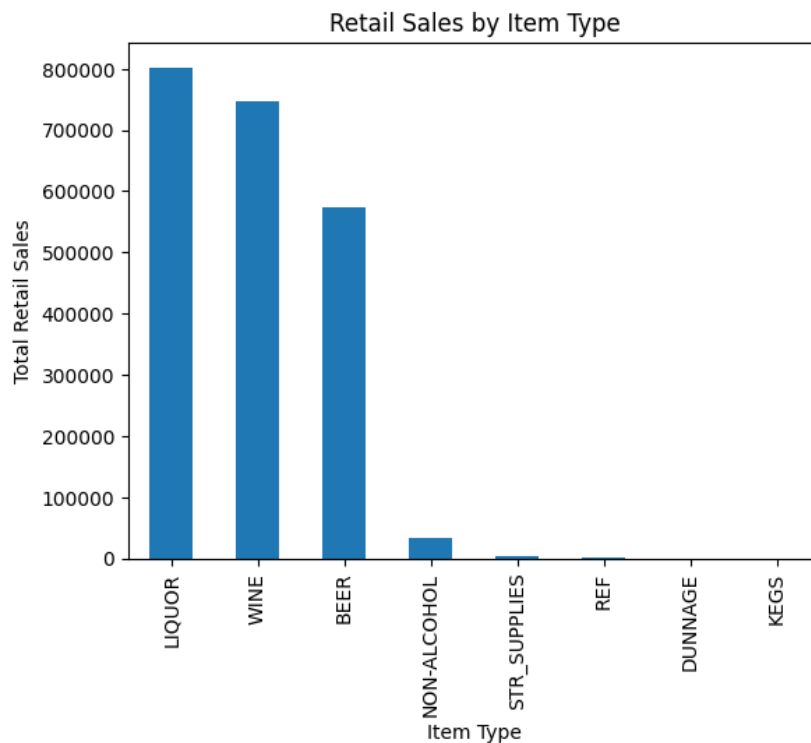


Both supply and demand are mostly directly proportional.

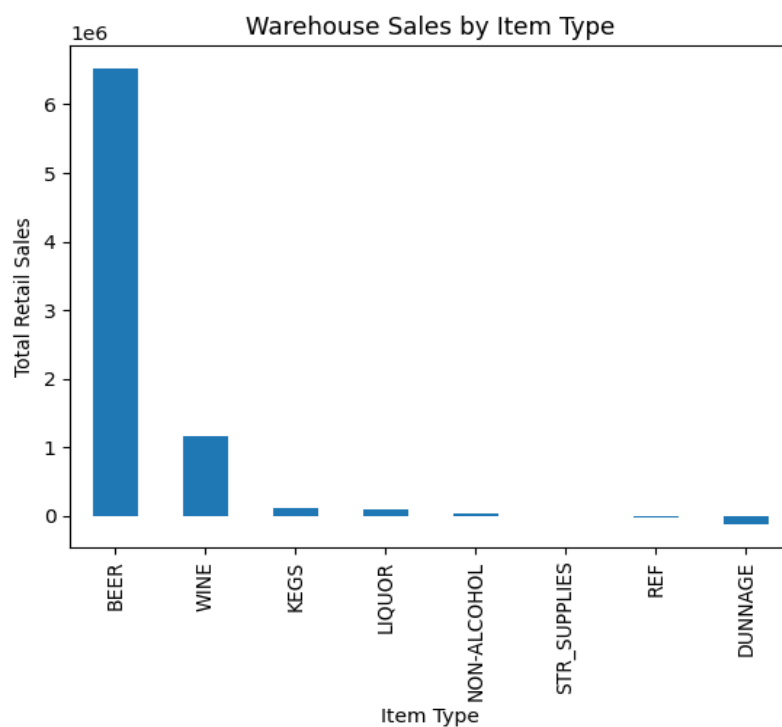
TASK-3

Item-Level Sales Trends

- Compare sales performance across different item categories.



Insight: Retail Sales of “Liquor” is highest followed by “Wine” and “Beer”. Least sales is for “KEGS”



Insight: Warehouse Sales of “Beer” is highest followed by “Wine” and “KEGS”. Least sales is for “Dunnage”

- **Identify high-performing and underperforming segments.**

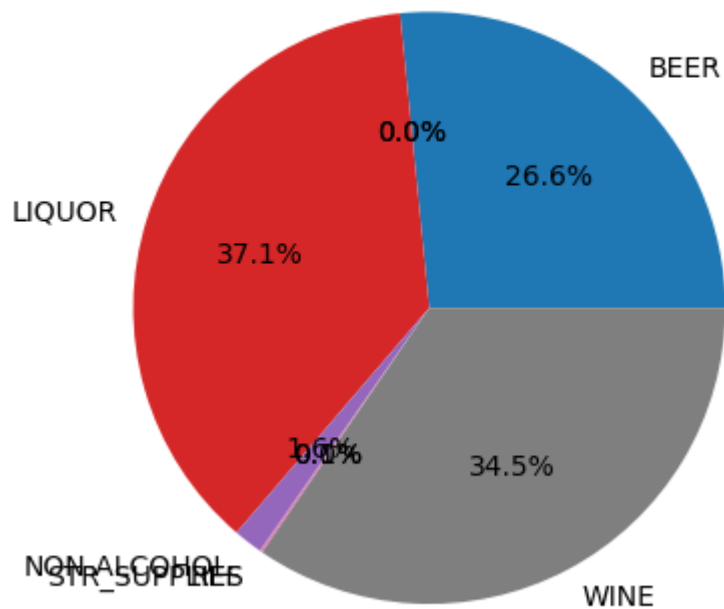
- High performing segments are : Liquor, Beer, Wine
- Low performing segments are : REF,Dunnage

TASK-4

Sales Breakdown by Category

- Present a categorical breakdown of total sales.

Sales Contribution by Category

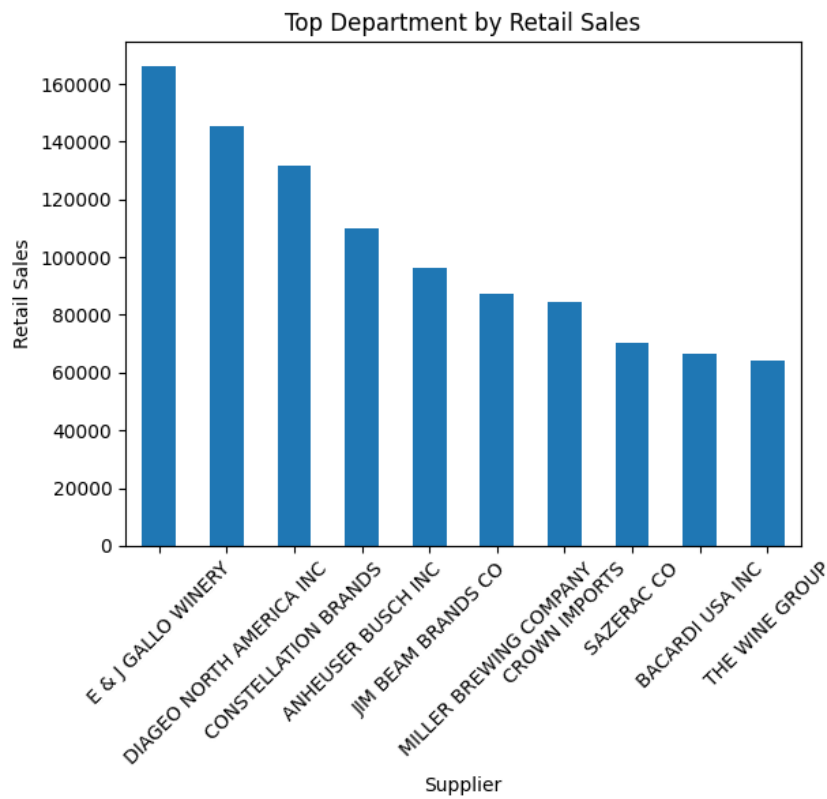


- Highlight the contribution of each category to overall revenue.
 - Liquor contribute maximum of 37.1% followed by Wine with 34.5%
 - Non-alcohol and str_supplies contribute very less to the overall sales

TASK-5

Sales Breakdown by Department

- Analyze department-wise sales distribution.



- Identify departments contributing the highest and lowest sales.

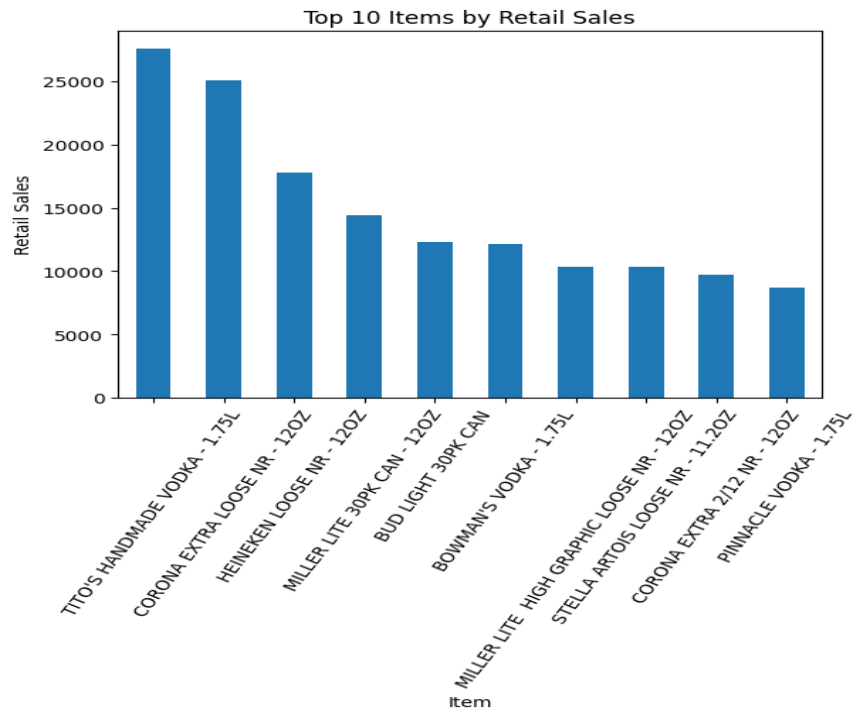
- Maximum sales is by E&J GALLO WINERY
- Lowest sales is by The wine group

TASK -6

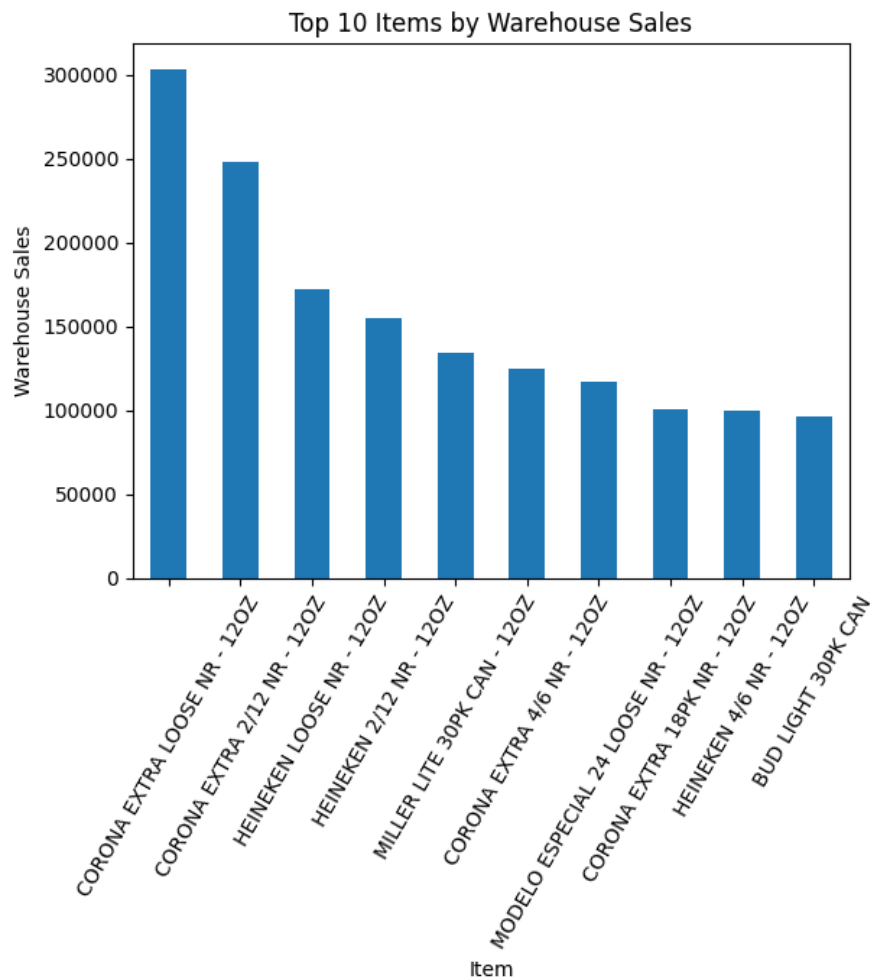
Top N Selling Items

- Identify the Top N products based on:

o Units sold



o Sales value



- **Interpret factors contributing to their performance.**

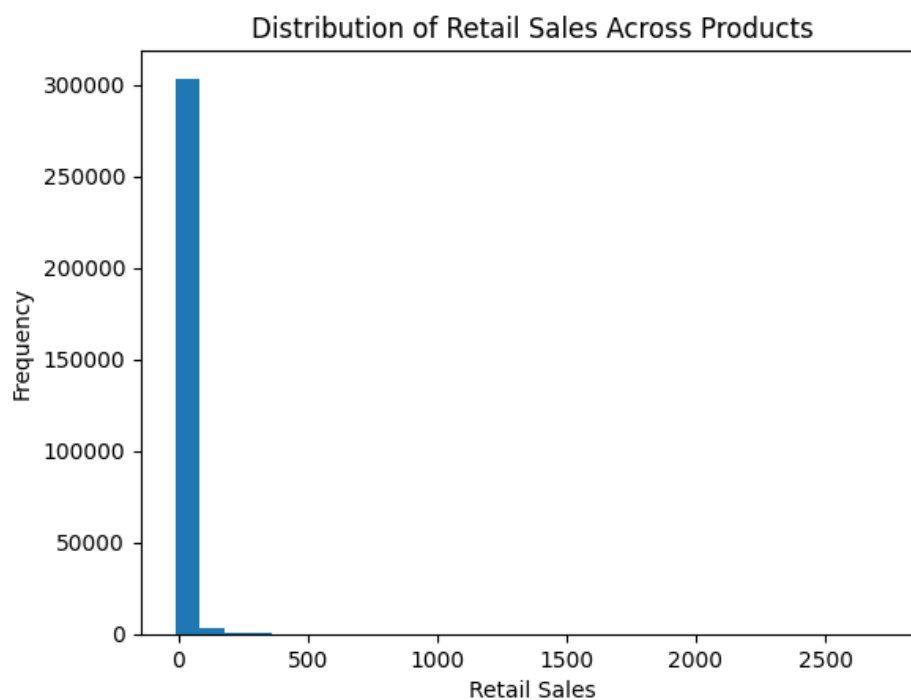
Several factors influence the performance of top-selling products:

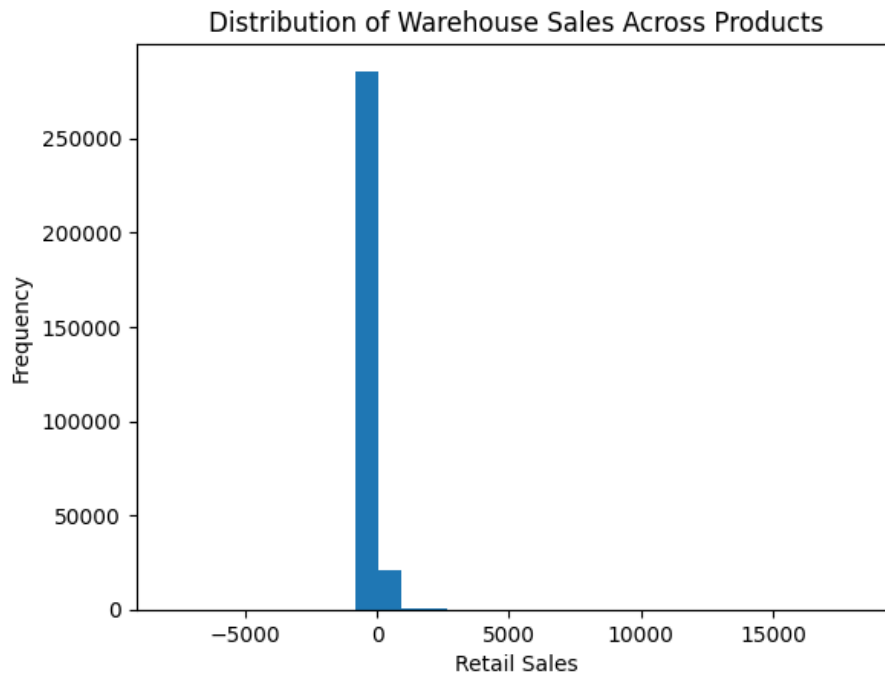
1. **Customer demand** – Popular products naturally achieve higher sales.
2. **Pricing strategy** – Competitive pricing can increase purchase frequency.
3. **Brand reputation** – Well-known brands often dominate sales.
4. **Seasonal demand** – Certain products may sell more during specific seasons.
5. **Distribution availability** – Products widely available across retail outlets tend to perform better.

TASK-7

Sales Distribution Analysis

- **Create histograms to analyse the distribution of sales across products.**





- Identify skewness, outliers, or concentration of sales among specific products.
 - The histogram analysis reveals that retail sales are not evenly distributed across products. Most products generate relatively low sales, while a few high-performing products contribute significantly to total revenue.
 - The distribution is positively skewed, indicating sales concentration among a small number of products. This suggests that businesses should focus on managing high-performing products effectively while improving the performance of lower-selling items.

TASK – 8

Warehouse v/s Retail Comparison

- **Develop side-by-side line or area charts comparing:**
 - o **Warehouse shipments**
 - o **Retail sales**



- Interpret inventory flow patterns and supply-demand gaps

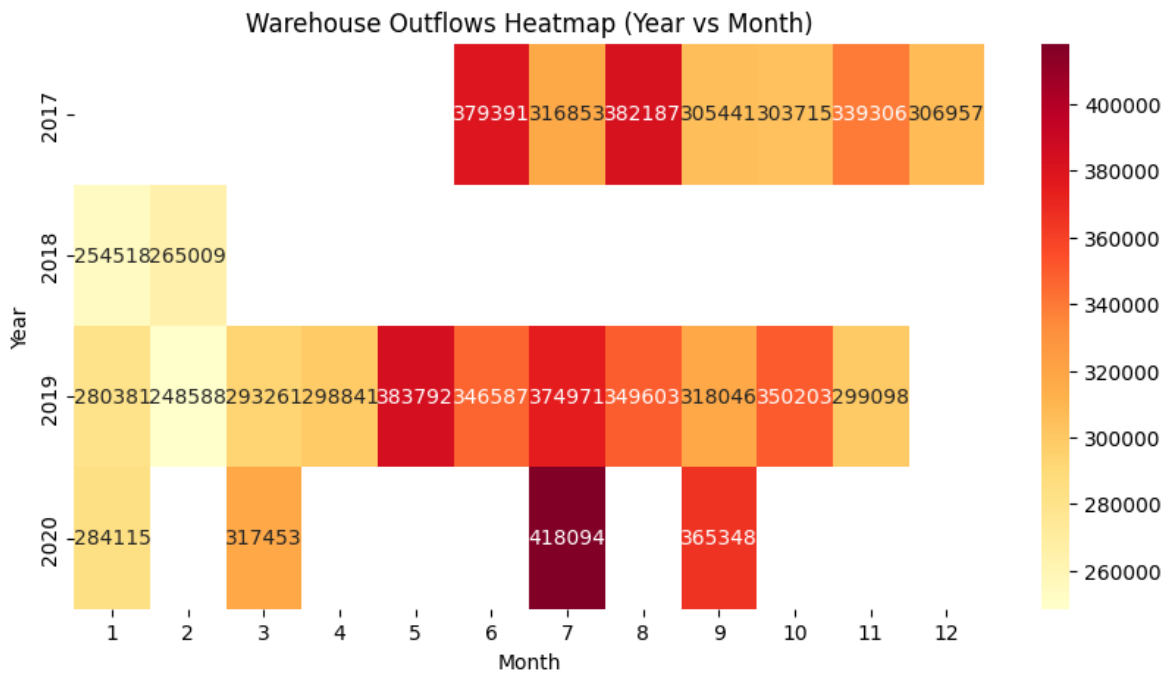
- The warehouse shipment trend generally follows the retail sales pattern, indicating that supply is aligned with consumer demand

TASK-9

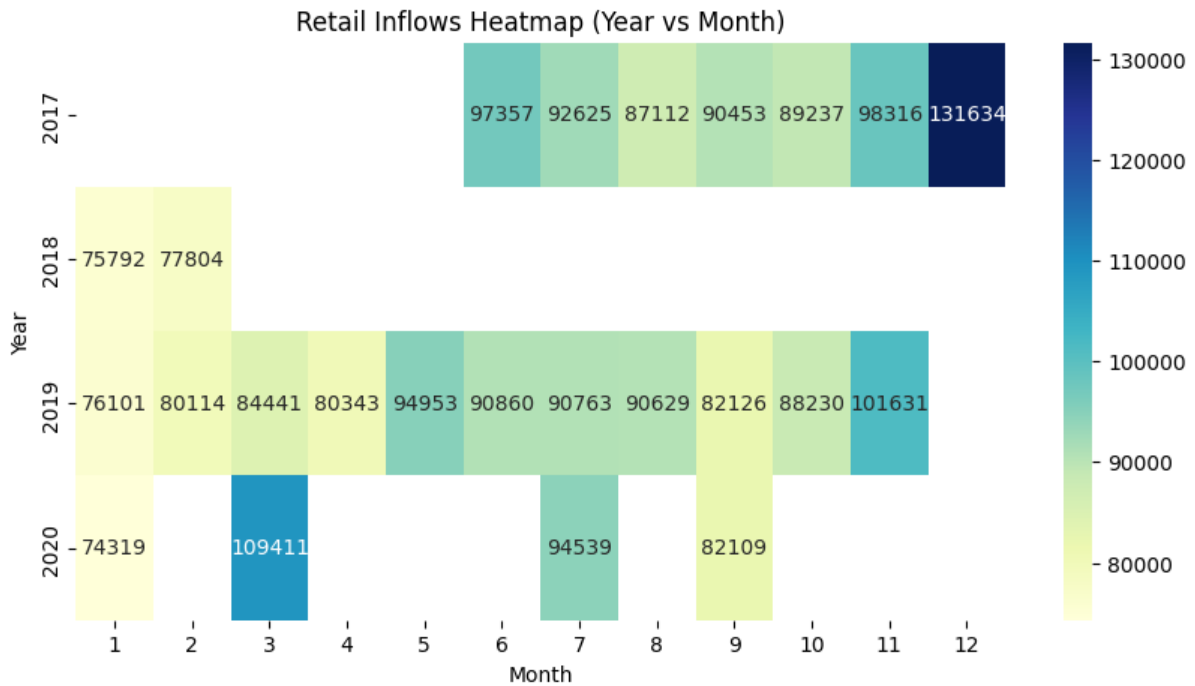
Heat map Analysis

- Create heat maps showing:

o Warehouse outflows over time



o Retail inflows over time



- **Identify peak periods and seasonal intensity.**

The heat map visualization provides a clear view of warehouse outflows and retail inflows across months and years. Darker regions indicate periods of high activity, revealing peak shipment and sales periods.

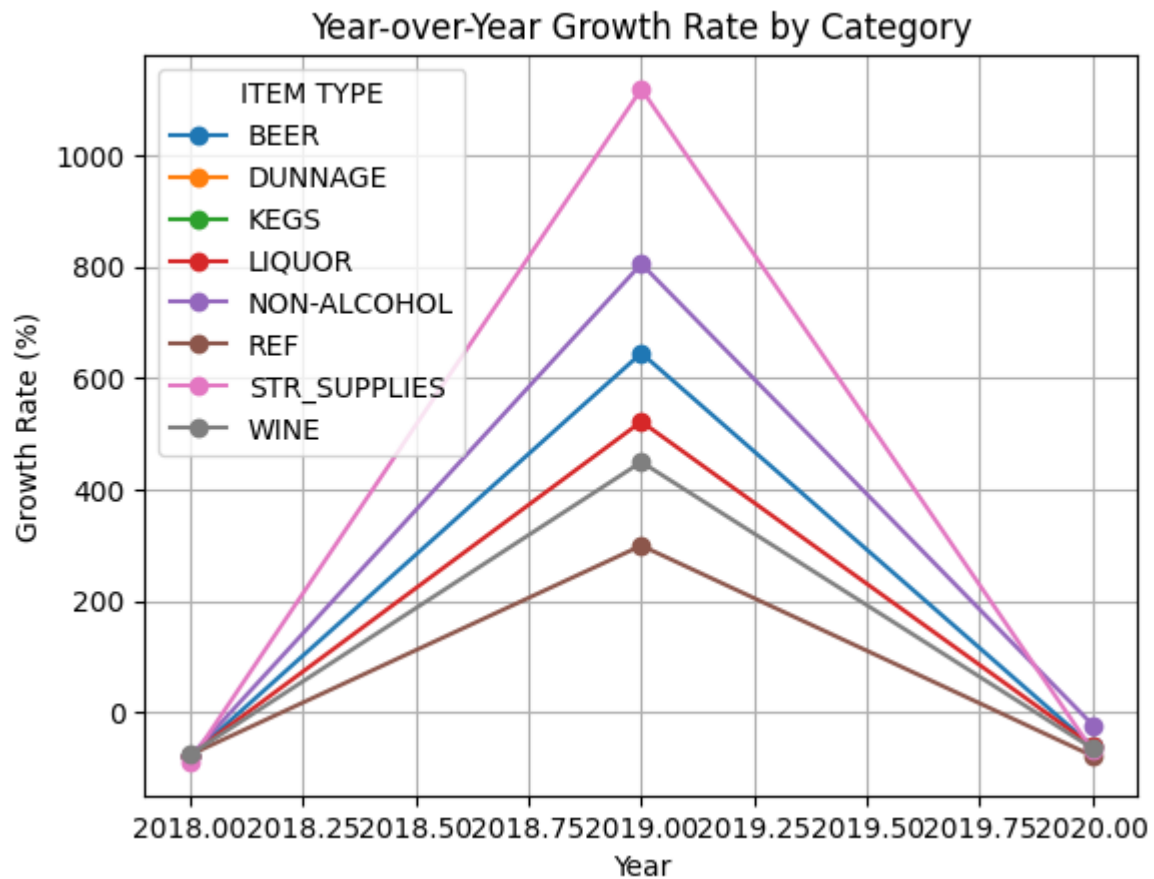
The heat map highlights peak warehouse shipments during mid-year months, particularly in months such as July and September.

Retail sales exhibit seasonal intensity during specific months across multiple years, suggesting consistent demand patterns driven by consumer purchasing behaviour.

TASK -10

Annual Growth Rate Analysis

- **Calculate and visualize year-over-year growth rates for each category.**



- **Identify fast-growing and declining segments.**

The year-over-year growth analysis reveals that **Non-Alcohol, Beer, Liquor, and Wine** categories experienced significant growth in 2019 compared to 2018, indicating strong market demand during that period. However, most categories recorded negative growth in 2020, suggesting a notable decline in sales. This pattern highlights fluctuations in consumer demand and emphasizes the importance of monitoring category performance to support strategic inventory and supply chain decisions.